Agentic Knowledge Sharing for Federated Collective Learning in Autonomous Vehicles

***Abstract*—Vehicles that learn only from their own observations cannot make a useful sign available to the rest of the fleet until those vehicles see it. Conventional federated averaging avoids uploading raw sensor data, but participating clients still exchange model updates to support one shared model. This paper presents an agentic knowledge-sharing framework for collective learning in traffic-sign recognition. YOLOv8n detects candidate sign regions, and each vehicle's own MobileNetV2 classifier marks a sign as known, rare, or new from its confidence. A reinforcement-learning agent decides whether to share or ignore an uncertain sign. Only an embedding from a common extractor leaves the vehicle; the image and the classifier weights stay local. The server accepts a package only when its best cosine similarity with a 43-class reference store reaches 0.70. In the prototype, verified packages are broadcast when broadcast is triggered manually, stored in each vehicle's temporary memory, and used on a later low-confidence classification. Local retraining later updates that vehicle's existing class layer. Classifiers use a 43-class output. The same single-sign crop set, drawn from 300 held-out German Traffic Sign Detection Benchmark scenes and covering 12 classes, was scored for three specialist vehicles. Classification accuracy moved from 35.16% to 81.32% for Vehicle A and from 17.58% to 79.35% for Vehicle B at Test 3, and from 39.56% to 73.63% for Vehicle C at Test 2, its last completed test. The combined sharing and retraining cycle raised accuracy for the three vehicles. A live CARLA run demonstrated the pipeline.**

***Index Terms*—Agentic AI, Federated Learning, Knowledge Sharing, Autonomous Vehicles, Traffic Sign Recognition, Reinforcement Learning**

I. Introduction

Autonomous vehicles rely on machine learning models to recognise traffic signs. A sign met by one vehicle may not appear in the view of the others. If each vehicle updates its model only from its own observations, that experience stays on the vehicle that made it. The rest of the fleet cannot use it until the same sign appears in their own view.

Federated learning has been studied as a way for connected vehicles to collaborate without sending raw sensor data [1], [7]. In conventional federated averaging, clients exchange model updates and the server returns one shared model [7]. That exchange does not make a sharing decision for each individual sign, and it does not keep a specialist classifier. Surveys of the approach for connected vehicles record the remaining difficulties: communication cost, non-identical data, and aggregation as the usual result [1]. Selective participation chooses which vehicle may upload [2]. Knowledge distillation can avoid the raw image [10]. Both still feed a shared model.

Reinforcement learning has been combined with federated learning for decision tasks in autonomous driving [2], [4]. Those methods typically aggregate a policy or a value model. They do not decide whether one observed sign is worth sharing.

This paper proposes an agentic knowledge-sharing framework for road signs. A shared detector finds a sign region. The vehicle classifier marks the crop as known, rare, or new from its confidence. A known sign stays local. A rare or new sign is checked against temporary memory. If there is no match, a Deep Q-Network agent decides whether to share it. A shared package contains an embedding and metadata. The server accepts it only when cosine similarity with a 43-class reference store reaches 0.70. In the prototype, broadcast is initiated manually. Accepted knowledge is held in each vehicle's temporary memory and can be used on a later low-confidence classification. Retraining later updates that vehicle's class layer, with the base layers frozen. Shared packages do not contain the camera image or the classifier weights.

The main contributions of this work are as follows:

- A sign-level share-or-ignore decision, made by a reinforcement-learning agent, rather than a decision only on which vehicle may upload.
- A server check that accepts an embedding only when its best match with a 43-class reference store reaches 0.70, and that refuses an unmatched package.
- A temporary memory that holds verified packages for use on a later low-confidence classification, without updating the classifier at once.
- Local retraining that freezes the MobileNetV2 base and updates the existing 43-way class layer, so each vehicle keeps its own classifier.
- An evaluation on the same 12-class single-sign crop set for three specialist vehicles, with a separate CARLA demonstration of the path.

II. Related Work

Federated learning for connected and automated vehicles is surveyed by Chellapandi et al. [1]. Most methods still follow the parameter aggregation of McMahan et al. [7], in which clients share model updates rather than raw data. Privacy is improved relative to centralised image upload, but communication remains costly and the usual result is one average model.

Perception work has applied the same pattern. Quéméneur and Cherkaoui propose FedPylot for real-time object detection in the Internet of Vehicles, using federated optimisation of YOLO models [3]. Zhang et al. study federated learning for cooperative perception and address heterogeneity with dynamic client weighting [12]. These approaches improve collaborative perception, but the object that leaves the vehicle is still a model update.

Other works have focused on knowledge sharing in vehicular networks. Yuan et al. propose federated broad learning for privacy-preserved knowledge sharing in the Internet of Vehicles [11]. Liang et al. share intermediate representations and soft predictions for data-free distillation [10]. The received item still supports a federated model.

Selective strategies have also been studied. Fu et al. choose which vehicle may upload, using reputation and model quality [2]. That is a control on participation, not a decision on the sign. Qi et al. survey federated reinforcement learning in which a policy or a value model is aggregated [4]. The agent in this paper does not upload a policy. It chooses share or ignore for one sign.

Traffic sign recognition has a long history. Sermanet and LeCun showed that a multi-scale convolutional network can name a cropped sign at high accuracy [6]. Houben et al. introduced the German Traffic Sign Detection Benchmark, which supplies the road scenes used here [5]. Continual learning remains relevant when a later update can disturb an earlier class. Kirkpatrick et al. and Li and Hoiem discuss this problem [8], [9]. They support a controlled update. They do not prescribe freezing the base of the classifier used here.

In summary, existing research has advanced federated learning, knowledge distillation, and selective participation for autonomous driving. However, most approaches still focus on exchanging model parameters or continuous aggregation. The framework proposed in this paper differs by allowing each vehicle to act as an agent that selectively shares compact and verified knowledge about rare traffic signs, supporting both immediate reuse from temporary memory and a later update of the vehicle's own class layer.

III. Methodology

This section presents the proposed agentic knowledge sharing framework for federated collective learning among autonomous vehicles. The system enables vehicles to selectively share useful knowledge about rare traffic signs instead of continuously exchanging full model parameters. The overall architecture consists of a local vehicle agent and a global knowledge aggregator, as illustrated in Fig. 1.

A. System Overview

Three vehicles take part. Vehicle A is specialised in prohibitory signs, Vehicle B in mandatory signs, and Vehicle C in danger signs. Each keeps its own classifier. One YOLOv8n detector is shared. It finds a sign region and does not name the class. A known sign stays local. A rare or new sign is checked against that vehicle's temporary memory. If a match

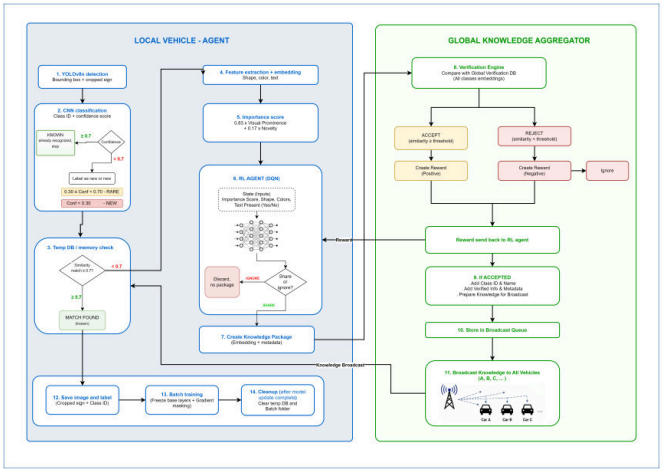


Fig. 1. Overview of the proposed agentic knowledge sharing framework.

is found, the cropped sign image is saved and the share path stops. If no match is found, the agent decides whether to share. The server accepts or rejects the embedding. An unmatched package is not broadcast. In the prototype, accepted knowledge is broadcast when broadcast is triggered manually. A later match can save the crop, and the vehicle then updates its existing class layer.

B. Local Detection and Classification

Traffic signs are first detected using YOLOv8n, which provides bounding boxes and cropped sign regions. The cropped images are then classified by that vehicle's MobileNetV2 network. The network has a 43-way class layer from the start. Based on the classification confidence, each detection is assigned one of three labels:

- **KNOWN**: confidence ≥ 0.70. The path stops.
- **RARE**: $0.30 \leq$ confidence < 0.70.
- **NEW**: confidence < 0.30.

These labels are confidence bands. A low score means the classifier is uncertain. It still returns a prediction. It does not prove that the sign shape is unseen. Only rare and new detections continue.

C. Temporary Database Check

Before the agent is called, a rare or new crop is compared with the vehicle's temporary memory. Cosine similarity is computed on embeddings from the same extractor. If a cosine similarity match of at least 0.70 is found in the local temporary memory, the vehicle retrieves the server-verified class label associated with the matching embedding. The locally observed traffic sign crop is saved for subsequent local retraining, and the DQN-based sharing decision is bypassed. This allows the vehicle to reuse previously shared knowledge without immediately modifying its CNN classifier. Below 0.70, the crop continues. A related package can fail this match.

D. Feature Extraction, Embedding, and Importance Scoring

When no match is found in the temporary database, visual features (shape, dominant colors, and text presence) and an embedding from a common extractor are extracted. An importance score is then calculated to quantify how valuable the observed knowledge is for sharing. The importance score is defined as

$$I(s) = \alpha \cdot V(s) + \beta \cdot N(s) \tag{1}$$

where $V(s)$ represents the visual prominence of the detected sign (based on the relative size of the bounding box), and $N(s)$ represents the novelty of the detection. The weights $\alpha = 0.83$ and $\beta = 0.17$ were selected by an offline search. Every pair from 0.00/1.00 to 1.00/0.00 was tested in steps of 0.01, giving 101 combinations, on 391 validation signs that were not taken from the 300 evaluation images. Novelty is a numerical input to this score. Its exact mapping is the one in the weight-search implementation. This score, together with the extracted features, forms the input state for the reinforcement learning agent. It is not the server's accept rule.

E. Agentic Decision Making

A Deep Q-Network (DQN) agent receives an eight-feature state representation and selects one of two actions: SHARE or IGNORE. The CARLA-integrated prototype runs the pre-trained DQN in inference mode. Live inference does not update its weights. If the agent selects IGNORE, the process ends for that detection. If SHARE is selected, a knowledge package is created. No controlled run showed that its choices are better than another sharing rule.

F. Knowledge Packaging

The knowledge package contains the embedding vector and metadata, including the predicted class, the confidence, and the source vehicle. Raw images and classifier weights are not included in the package. Leaving the image local reduces exposure. It does not prove that an embedding cannot carry information about the crop.

G. Global Verification and Reward

The knowledge package is sent to the global verification engine, which compares the embedding against a reference database of embeddings for the 43 sign classes. The reference embeddings were built with the same extractor. If the best match reaches cosine similarity 0.70, the knowledge is accepted and given that class identity; otherwise, it is rejected and is not broadcast. A corresponding positive or negative reward is generated and sent back to the local RL agent as feedback for a submitted package only. An IGNORE decision is not verified. The 0.70 score is a reference match. It is not proof that the class is correct, and it is not a security or authenticity guarantee.

H. Broadcasting

Accepted knowledge packages are placed in a server-side broadcast queue. Broadcasting is manually triggered during the implemented workflow, after which queued packages are distributed to all three participating vehicles, including the originating vehicle. Each receiving vehicle stores the verified embedding and associated metadata in its local temporary memory for use during subsequent uncertain traffic sign detections. Receiving a package does not immediately update the local CNN weights.

I. Local Batch Retraining

Local retraining uses traffic sign crops observed and stored by the vehicle following successful temporary-memory matches. During retraining, the MobileNetV2 feature extraction base remains frozen, while the existing 43-class Dense classification head is updated using gradient masking. After processing, the stored training crops are removed from their temporary class-organised folders. This approach aims to incorporate shared knowledge into the local classifier while limiting changes to the pretrained feature extractor. However, the experiments do not establish that catastrophic forgetting is eliminated. No fixed clear period is claimed.

J. Pipeline Summary

The complete processing flow can be summarized as follows: Detection → Classification → Temporary Database Check → Feature and Embedding Extraction → Importance Scoring → RL Decision → Knowledge Packaging → Global Verification → Manually Triggered Broadcast → Local Reuse → Batch Retraining. This design supports selective, verified knowledge sharing among autonomous vehicles. Model weights are not averaged.

IV. Experimental Setup and Results

This section describes the experimental setup and presents the evaluation results of the proposed agentic knowledge sharing framework.

A. Dataset and Experimental Setup

The scenes are taken from the German Traffic Sign Detection Benchmark (GTSDB) [5]. That benchmark has 900 road scenes: 600 training images and 300 evaluation images. The 600 training images were used to prepare the detector and the sign crops for classifier training. The 300 evaluation images were kept separate and form the held-out set. Sign regions were taken from these scenes. A scene can contain more than one sign, so training and testing use the cropped sign, not the whole scene. The label list runs from 0 to 42 and follows the recognition taxonomy associated with the German Traffic Sign Recognition Benchmark [6].

Three specialized local models were trained:

- **Vehicle A**: primarily trained on Prohibitory signs
- **Vehicle B**: primarily trained on Mandatory signs
- **Vehicle C**: primarily trained on Danger signs

Each classifier has a 43-way class layer from the start. Initial training is not spread evenly across that list. A representative subset of 12 classes covering all three categories was selected for evaluation. The selected classes are:

- **Prohibitory**: 0 (speed limit 20), 1 (speed limit 30), 4 (speed limit 70), 7 (speed limit 100)
- **Danger**: 18 (danger), 23 (slippery road), 27 (pedestrian crossing)
- **Mandatory**: 33 (go right), 35 (go straight), 40 (round-about)
- **Other**: 13 (give way), 14 (stop)

The scored test uses single-sign crops from the 300 held-out evaluation images. About 8–10 crops were scored for each class. The counts are approximate and are taken from the test logs. These crops were not used to train the classifiers or to choose the importance weights. All three classifiers retain 43 output classes, even though only 12 classes are included in the reported evaluation. A separate live run in CARLA Town01 validates detection, classification, the share decision, verification, and broadcast. It is an integration demonstration, not an accuracy experiment.

B. Evaluation Protocol

Each vehicle was evaluated in multiple stages on the same crop set:

- **Test 0**: Original local model (before knowledge sharing)
- **Test 1**: After the first round of knowledge sharing and local retraining
- **Test 2**: After the second round of knowledge sharing and local retraining
- **Test 3**: Final evaluation after additional adaptation

Vehicle A and Vehicle B were evaluated across all four stages (Test 0 to Test 3). Vehicle C was evaluated across three stages (Test 0 to Test 2). Performance was measured as classification accuracy: the share of single-sign crops assigned the correct class. Overall accuracy uses all scored crops for that vehicle. Class accuracy uses the crops of one class.

C. Overall Performance

Table I summarizes the overall accuracy of the three vehicles across different test stages. From Test 0 to the last completed test, Vehicle A rose by 46.16 percentage points, Vehicle B by 61.77 percentage points, and Vehicle C by 34.07 percentage points. These are percentage-point changes, not percentage increases. Vehicle C's final reported stage is Test 2. The largest step for each vehicle is from Test 0 to Test 1. Later tests still change the figure, and not always upward. The rise is the measured change after the combined cycle. A no-sharing retrain was not run, so the table does not separate sharing from retraining, and it does not isolate the agent or the server check.

TABLE I
OVERALL ACCURACY OF VEHICLES ACROSS TEST STAGES

Vehicle	Test 0	Test 1	Test 2	Test 3
Vehicle A	35.16%	69.23%	72.53%	**81.32%**
Vehicle B	17.58%	67.03%	64.13%	**79.35%**
Vehicle C	39.56%	74.73%	73.63%	–

D. Class-wise Results

Tables II, III, and IV present the class-wise accuracy for each vehicle.

TABLE II
CLASS-WISE ACCURACY OF VEHICLE A

Class	Test 0	Test 1	Test 2	Test 3
0	87.50%	87.50%	75.00%	87.50%
1	28.57%	57.14%	71.43%	57.14%
4	37.50%	87.50%	87.50%	87.50%
7	80.00%	70.00%	70.00%	90.00%
13	0.00%	100%	100%	87.50%
14	22.22%	100%	100%	77.78%
18	0.00%	80.00%	80.00%	100%
23	0.00%	77.78%	88.89%	66.67%
27	100%	0.00%	0.00%	85.71%
33	0.00%	16.67%	33.33%	66.67%
35	60.00%	60.00%	60.00%	60.00%
40	0.00%	50.00%	75.00%	100%

TABLE III
CLASS-WISE ACCURACY OF VEHICLE B

Class	Test 0	Test 1	Test 2	Test 3
0	25.00%	75.00%	75.00%	75.00%
1	14.29%	42.86%	42.86%	71.43%
4	0.00%	50.00%	50.00%	50.00%
7	20.00%	60.00%	70.00%	70.00%
13	0.00%	100%	0.00%	100%
14	22.22%	88.89%	88.89%	88.89%
18	0.00%	70.00%	100%	100%
23	0.00%	77.78%	77.78%	66.67%
27	28.57%	85.71%	28.57%	100%
33	0.00%	0.00%	71.43%	71.43%
35	80.00%	80.00%	80.00%	80.00%
40	75.00%	50.00%	75.00%	75.00%

TABLE IV
CLASS-WISE ACCURACY OF VEHICLE C

Class	Test 0	Test 1	Test 2
0	0.00%	50.00%	75.00%
1	100%	42.86%	85.71%
4	0.00%	75.00%	50.00%
7	20.00%	70.00%	20.00%
13	25.00%	100%	100%
14	55.56%	88.89%	77.78%
18	70.00%	100%	90.00%
23	55.56%	88.89%	88.89%
27	85.71%	57.14%	100%
33	0.00%	33.33%	66.67%
35	40.00%	100%	60.00%
40	0.00%	75.00%	75.00%

E. Analysis

The observed improvements are consistent with the benefits of the integrated knowledge-sharing and local adaptation workflow. However, because the evaluation did not include controlled ablation experiments, the individual contributions of the DQN, temporary-memory matching, server verification, and retraining cannot be isolated. The results therefore demonstrate improvements across the evaluated system stages rather than establishing the causal effect of any single component.

Several classes that initially had near-zero accuracy (such as give way, danger, and roundabout) were named at the last completed test. Temporary performance drops were observed on some classes. Class 27 on Vehicle A was 100% at Test 0, 0% at Test 1 and Test 2, and 85.71% at Test 3. That decline is consistent with interference during later updates. The experiment did not isolate the cause, and the later rise does not establish a recovery mechanism. Freezing the base layers and gradient masking limit the scope of the update. They do not guarantee that every class stays flat.

V. Discussion

The experimental results show that the proposed agentic knowledge sharing framework can reduce the initial category bias of specialized vehicles on the scored crops. By allowing vehicles to selectively share verified knowledge about rare traffic signs, the system supports collective learning without the continuous exchange of full model parameters. That is a different architecture from conventional averaging [7] and vehicle selection [2]. It is not a measured ranking against those methods.

The temporary memory and the later class-layer update play different roles. Storage does not change the classifier. A later match can save a crop, and retraining then updates the class layer. The reported accuracy is from the classifier after retraining, not from the memory match alone. Freezing the MobileNetV2 base and updating the Dense head with gradient masking limit the scope of the update. The class-27 drop shows that this does not guarantee every class stays flat.

The evaluation has limits. Only 12 classes were scored, with about 8 to 10 crops for each class, because the 300 held-out evaluation scenes did not provide enough single-sign crops for every class. One crop can move a class percentage by several points. Each stage is one completed run, so no interval is reported. Vehicle C was not run to Test 3. No ablation and no averaging baseline were run. The absence of the raw image was not measured as a bandwidth saving. The 0.70 threshold was not scored for false acceptance or false rejection. Broadcast was initiated manually, so automatic scheduling and latency were not evaluated. The CARLA Town01 run checks the path. It does not produce the accuracy figures, and it does not add a result under weather, camera, or fleet variation.

These limits bound the claim. They do not remove the rise in Table I. The rise is on the 12-class crops that were scored, after the sharing and retraining cycle. It is not a result for every sign class, and it is not a result from the simulator.

VI. Conclusion

This paper presented an agentic knowledge sharing framework for federated collective learning among autonomous vehicles. Unlike conventional federated averaging, which exchanges model updates to support one shared model, the proposed system lets a vehicle decide whether one uncertain sign is shared, checks the embedding against a reference store, and lets a receiving vehicle update its own class layer.

The framework combines local detection and classification, embedding-based similarity checking, importance scoring, reinforcement learning-based decision making, global verification, and controlled local retraining. On held-out single-sign crops, Vehicle A moved from 35.16% at Test 0 to 81.32% at Test 3, Vehicle B from 17.58% to 79.35% at Test 3, and Vehicle C from 39.56% to 73.63% at Test 2. The CARLA run showed the path from detection to broadcast. The study does not show that the agent or the verification step alone caused the gain, and it does not show a result for all 43 classes. It does show that selective sharing, a check before broadcast, and an update of the vehicle's own class layer can be run together.

Future work should collect more single-sign crops, run Vehicle C to Test 3, separate the steps that were run together, place an averaging baseline on the same 12 classes, score the 0.70 threshold for false acceptance and false rejection, and evaluate broadcast timing and package size. An automatic broadcast rule was not implemented in this prototype.

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